Seven architectures, one rung

N=48, degree 5, 4 colours, leak-free (--shuffle-adj --fall) | 67 cells, 21 complete rungs, 3 seeds each

THE OBSERVATION

7 complete rungs sit at rho_50 0.740-0.765, spanning 1,746,490 to 4,899,898 trainable parameters -- a factor of 2.8 -- across six different interventions: shallower depth, a narrower residual stream, a narrower MLP, a linearised MLP, a frozen random MLP, and no MLP at all. Removing the entire MLP and halving the depth put the model in the same place.

IT IS NOT A SQUINT

Across-rung spread is 0.0280 in mean_solve; the worst ACROSS-SEED spread within a single one of these rungs is 0.2918 -- exactly the pooled s* this programme froze as its yardstick, because s* was defined as the largest across-seed spread in the grid. The pile-up spans one tenth of the bar every verdict here is scored against. Shuffle which rung each cell belongs to, 20,000 times: p = 0.569.

AND THE HONEST LIMIT, WHICH IS THE POINT OF THE PAGE

p = 0.569 is also what a test with no power returns. The same test on the full model's depth ladder rejects at p = 0.045 -- but it took that ladder's entire dynamic range, a 0.736 separation, to get there at 3 seeds, and the band's is 26x smaller. The claim is NO DETECTABLE SEPARATION, resolution limit between 0.028 and 0.736. Not equivalence; a 3-seed grid cannot support that.

WHAT IT CANNOT SETTLE, AND WHAT IS RUNNING

(A) BARRIER: 0.76 is a floor any incomplete architecture hits however much capacity it is given back. (B) CAPACITY: the ablations merely cost about the same at this budget. The grid cannot decide, for a reason invisible row by row -- every ablated rung in it was run at AT MOST the hub's depth. Stages D (width) and E (depth) give one more; last page.

rho_50 = realized hint density at which solve rate crosses 0.5 on a 14-point x 400-instance grid, LOWER IS BETTER, reported as >0.95 when it never crosses. mean_solve = unweighted mean over the 14 bins, which never censors. Every cell trained and evaluated with --shuffle-adj.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND      N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND      N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND      N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND      N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

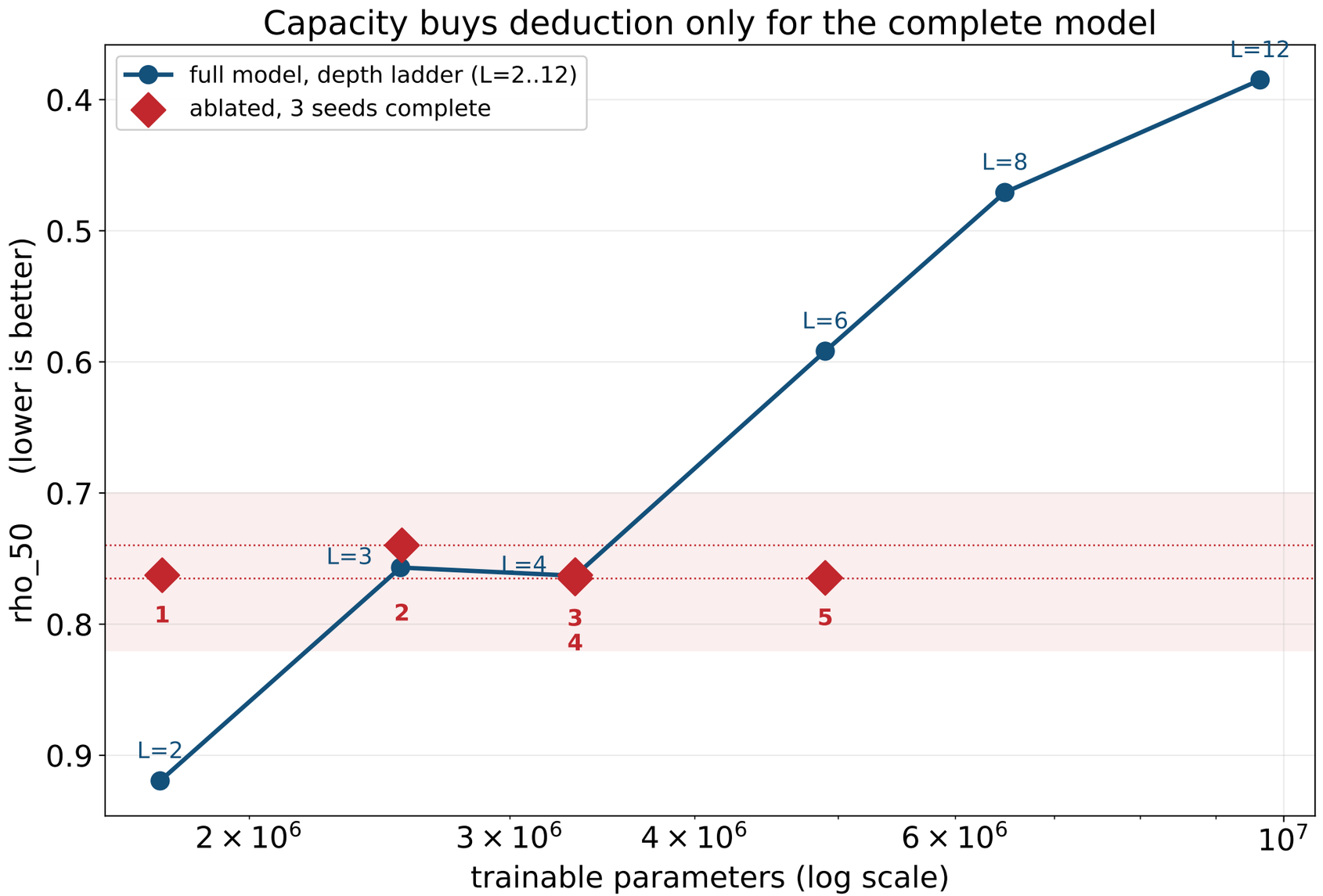
```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. Every architecture on page 1 has to produce this token by token, with no scratchpad and no tool. rho_50 asks how few hints it can be given and still finish.

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.



THE ABLATED RUNGS, numbered by size

	rho_50	trainable		
1	0.763	1,746,490	L6	no_mlp
2	0.740	2,535,994	L6	full ff256
3	0.765	3,320,378	L4	linear
4	0.763	3,320,890	L6	rf_mlp
5	0.765	4,899,898	L6	linear

band width 0.0253 in rho_50, over a 2.8x range in parameters

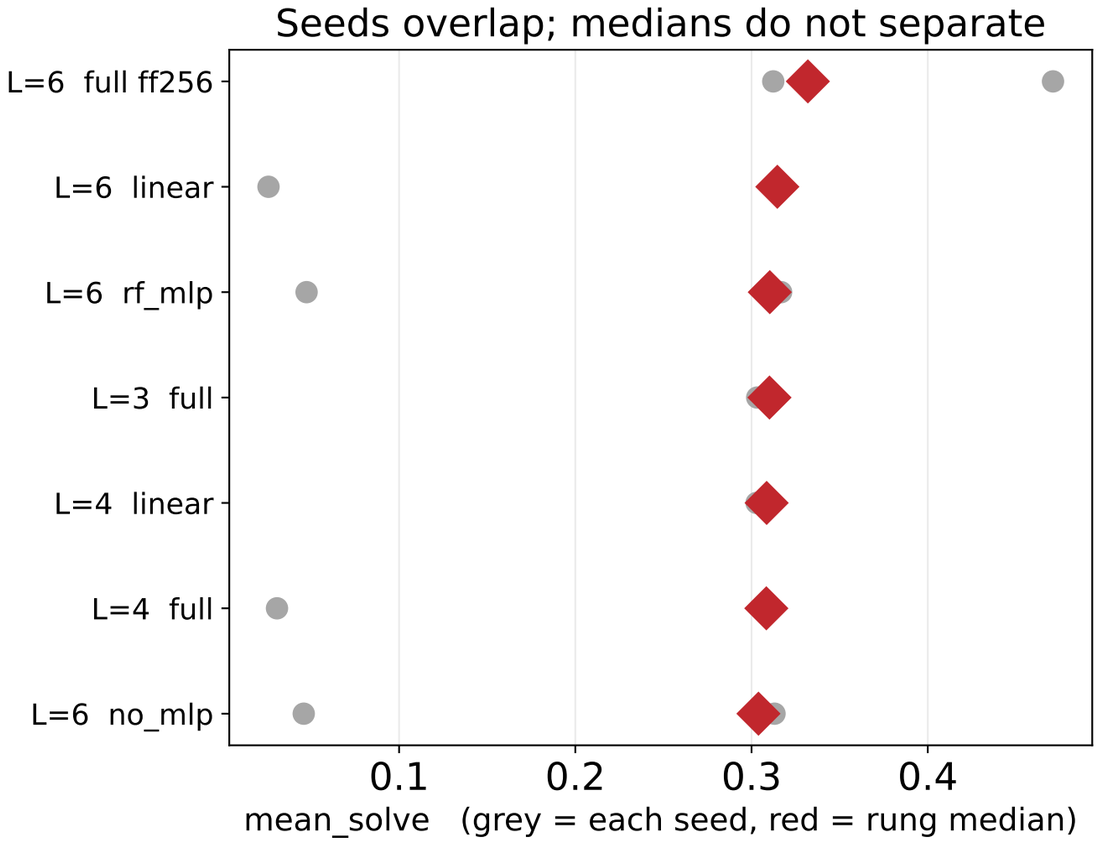
THE FULL MODEL passes straight through the band at L=3-4 and keeps descending: 0.592 at L=6, 0.471 at L=8, 0.385 at L=12.

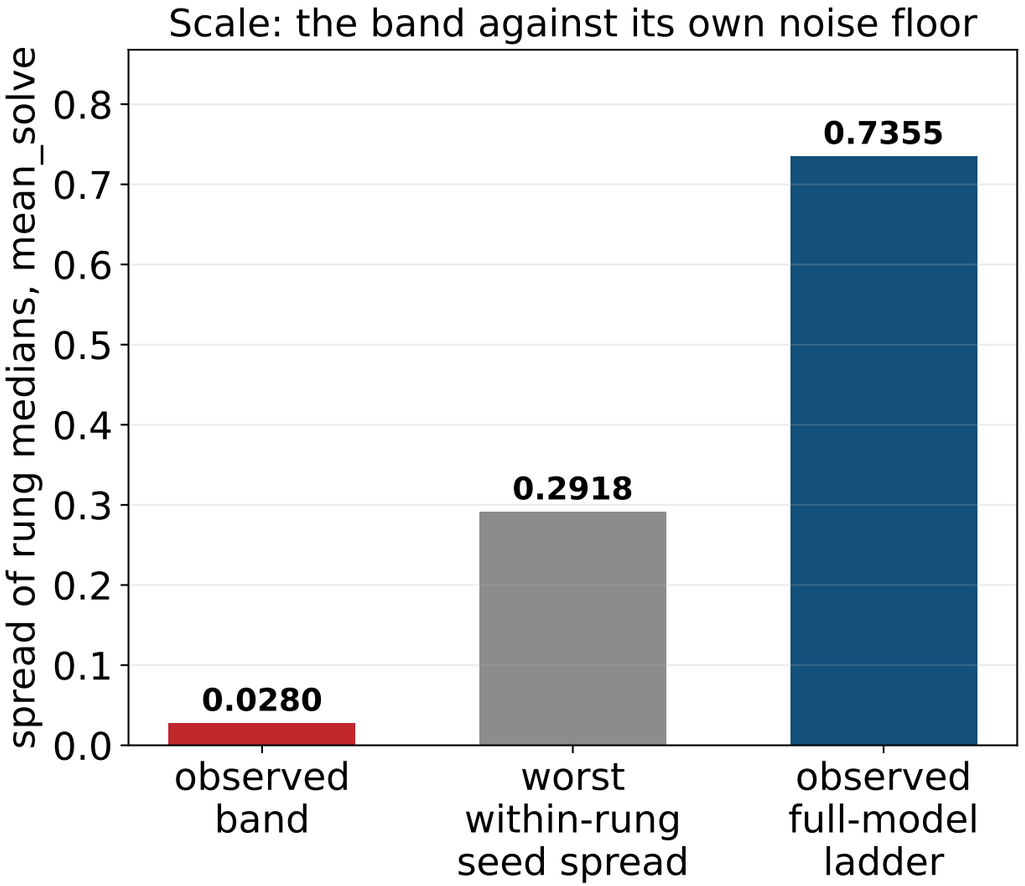
THE ABLATED MODELS stop there, incl. ones with MORE parameters than full L=3. And no_mlp d256 has 1,746,490 and SOLVES at 0.763, while full d128 has 2,059,834 -- 18% MORE -- and never crosses 0.5 at any of 14 densities on any of 3 seeds.

Censored rungs (rho_50 never crosses 0.5) cannot be placed on this axis and are omitted: full d128, no_mlp d128, rf_mlp L=4, rf_all, nhead=1 at both depths, full L=1, full d64. All of them sit BELOW the band, so omitting them understates the collapse rather than flattering it. Commit 9b8e140.

The same test, run twice: on the band and on something known to differ

Permutation test on mean_solve: shuffle which rung each cell belongs to, keep group sizes, recompute the spread of rung medians





band: p = 0.569 (not rejected) **full-model depth ladder: p = 0.045 (rejected)**

The positive control is not decoration. p = 0.569 on the band is also what a test with no power returns, so the identical test runs on rungs the whole programme treats as real. It rejects -- but only at p = 0.045, and only on a 0.736 separation, 26x the band's. The honest reading is a resolution limit, not an equivalence. 20,000 shuffles, RNG seed 0.

Stage C: four registered rows, scored against a bar frozen before they ran

51114734, 13 of 15 cells done. Paired control = stage A's L=6 d256 ff1024 h4 full: rho_50 0.592, mean_solve 0.5389

P1 FALSIFIED -- the residual stream width is load-bearing, and this one is clean

full at d_model 128 is censored 3/3 at every one of 14 hint densities, mean_solve 0.0273 [spread 0.0036] at 2,059,834 trainable. 2 of 3 seeds got BELOW the 1.64 plateau, so unlike every other decisive failure in this grid the trainability caveat does not touch it. Findings 13 and 26 said depth beats width; at N=48 they are wrong about width. Removing the ENTIRE MLP at d256 costs 0.235 of mean_solve; halving the stream costs 0.512.

P2 UNSCOREABLE AS REGISTERED, and recorded as that rather than as a tie

Its quantity is a difference of differences and both d128 terms are censored, so it does not exist. The mean_solve fallback reads 0.0291 against 0.0273 -- two cells on the floor, the 'both cells failed' trap -- and all three no_mlp d128 seeds are above the plateau too. d128 is below the width this task needs, and nothing about the MLP can be read at a width where the full model fails as well.

P3 PASSES THE REGISTERED RULE BY 0.0004 AND NOTHING MORE

ff 1024->256: 3/3 crossed, rho_50 median 0.740, where the free/not-free threshold is 0.740. That is a tie with the bar, not a demonstration that quartering ff is free. Read mean_solve, where there is no boundary: 0.3320 against 0.5389 is -0.2070, essentially what removing the entire MLP costs (-0.2350). A 4x narrower MLP is worth about what no MLP is worth -- so the 'all' in 'the MLP is all-or-nothing' is its WIDTH.

P4 IS A RESULT, NOT A FAILED CELL -- and it closes stage B's open question

nhead=1 with warmup 5 escapes the plateau 0/3 (1.648/1.648/1.647), mean_solve 0.0257 [0.0048] -- matching stage B's N5 at warmup 0 (also 0/3, 1.657/1.648/1.650, 0.0270). Two independent optimizer settings failing the same way is enough to write N5's 'WORSE' as a property of having ONE HEAD rather than of the schedule, which is precisely what stage B could not say. P5 (d64) is at 1 of 3 seeds: censored, mean_solve 0.0011.

'Free' means rho_50 stays at or below 0.592 + s* = 0.740, with s* = 0.1484 frozen before stage C ran and not recomputed since. Stage C's rungs carry a non-default d_model, ff or warmup, so they keep their own keys and stay OUT of the pooled bar stages A and B were scored against. Commit 9b8e140.

What is running, and what each outcome would mean

Both jobs hand an ablated model MORE of a resource than the hub it is compared against -- which no cell in the grid has done

STAGE D, 51151957, THE WIDTH ARM -- 2 of 6 cells complete

D1 (full at d192, 3,381,562 trainable) has two seeds at rho_50 0.759 and 0.752, mean_solve 0.3138 and 0.3266, both escaping the plateau. Its registered mean_solve window [0.10, 0.45] holds, but its 'lands strictly between its neighbours' reading does not: d192 landed IN the band with everything else. Bisecting the width cliff produced another member of the pile-up. D2 (attention-only at d384, 3,799,354 trainable, 22% under the hub) has no completed cell yet.

STAGE E, 51171824, THE DEPTH ARM -- submitted today, 3 of 6 running

Attention-only at L=8 (2,274,874 trainable) and L=12 (3,331,642), three seeds each, everything else at the hub's settings. L=8 and L=12 were chosen because the full model is already measured at both, so each rung has a paired partner at identical depth and identical budget. Registered falsifier of the barrier reading: either rung with a median rho_50 below 0.740 on at least 2 of 3 crossing seeds. No partial credit, and the 0.740 tie-break is written down in advance this time.

THE PREDICTION THIS DECK IS ON RECORD FOR, MADE BEFORE D2 FINISHED

Stage E's header backs D2's FALSIFIER over D2's own prediction: under the barrier reading, width is not the currency either, so d384 attention-only should land beside d256 attention-only at 0.76 rather than reaching the hub. If D2 and both E rungs all stay at 0.76, the barrier is demonstrated on two independent axes. If D2 clears it while E does not, the currency is width specifically -- a sharper result than either job registered.

THE WAY THIS PRODUCES NOTHING, NAMED IN ADVANCE

Trainability, not capacity. Attention-only escapes the 1.64 plateau 2/3 at L=6 and 0/3 at L=4, so a DEEPER stack failing to escape is the EXPECTED failure mode and gets written as 'did not train in 375k steps', never as evidence for a barrier -- the trap P2 fell into. The budget is deliberately not raised, since a rung on a different budget cannot be read against this ladder.

Previous best on this eval, same 14-point grid and same budget: the hub at rho_50 0.592 / mean_solve 0.5389 and 4,899,898 trainable, and below it only full L=8 (0.471) and L=12 (0.385) at 6.5M and 9.6M. Nothing in this deck beats it, and nothing was expected to -- the brief is the smallest architecture that still learns direct deduction, so the standing minimum remains attention-only at d256, 1,746,490 trainable, rho_50 0.763. Commit 9b8e140.